

CARMEN ARMAS MONTALVO

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EDUCATION

Ph.D. in Applied Economics University of Minnesota	2022 – Present Minneapolis, USA
Master's Degree in Economics (<i>First in Cohort, 1/16</i>) Pontifical Catholic University of Peru	2014 – 2016 Lima, Peru
Bachelor's Degree in Economics Pontifical Catholic University of Peru	2008 – 2013 Lima, Peru
Specialization in Applied Statistics National Institute of Statistics and Informatics	2021 Lima, Peru

SKILLS

Technical skills: Econometric modeling and causal inference
Research fields: Labor economics, informality, worker sorting, and market allocation in developing economies.
Software and programming language: Stata, R, ArcGis, Qgis, Julia, Python, and LaTeX.
Languages: Spanish (Native), English (Fluent), and French (reading proficiency).

RESEARCH AND PROFESSIONAL EXPERIENCE

Hubert H. Humphrey School of Public Affairs, University of Minnesota Graduate Research Assistant - (Supervisor: Caroline Krafft)	Aug 2024–Present Minneapolis, Minnesota
• Co-author of a working paper on labor mismatch in Egypt; currently preparing for journal submission. • Contributed to research design to study how workers sort across sectors in response to wage differentials and skill mismatch. • Implemented econometric analysis and constructed measures of labor market allocation and mobility. • Examined informality dynamics using a job-ladder framework to analyze transitions.	
Department of Applied Economics, University of Minnesota Graduate Research Assistant (Supervisor: Joe Ritter)	Jan 2024 – Aug 2024 Minneapolis, Minnesota
• Implemented and evaluated a novel Difference-in-Differences estimator in R, incorporating cross-validation techniques. • Assessed estimator performance using open-access replication datasets. • Produced comparative analysis of labor force participation patterns across countries.	
Hubert H. Humphrey School of Public Affairs (UMN) Data Analyst - Direct Supervisor: Christina Ewig	Jun 2023 – Aug 2023 Minneapolis, Minnesota
• Reconstructed and standardized Stata workflows to analyze administrative and survey data for the 2024 Status of Women & Girls+ in Minnesota report. • Produced statistical analyses and validated key indicators for publication.	
Government of Peru Data and Policy Analyst in the Ministry of Labor and Employment Promotion	Dec 2017 – Aug 2022 Lima, Peru
• Contributed to the design of Peru's National Decent Work Policy, analyzing labor market frictions and employment quality. • Classified occupations using task-based typologies (routine vs. non-routine) to study skill allocation. • Developed a predictive model of job quality risk using LASSO and ROC analysis.	
Impact Evaluation Specialist in the Ministry of Education • Designed and implemented quasi-experimental evaluations using administrative education data.	

Group for the Analysis of Development (GRADE)

Jun 2014 – Sep 2017

Research Assistant

Lima, Peru

- Estimated rural poverty using small-area estimation techniques combining census and survey data.
- Applied spatial econometric models (SARAR, SARE) to analyze geographic spillovers.
- Evaluated the impact of tax exemptions on economic activity using satellite night-light data and household panel surveys.

CONFERENCES AND PRESENTATIONS

Mismatch in the Egyptian labor market

2026 Africa Meeting of the Econometric Society

Jun 2026

El Cairo, Egypt

The Impact of Mobile Phones on Working Conditions in Peru

XXIII RIDGE FORUM – Labor Workshop

May 2026

Cave Hill, Barbados

Did “Ni Una Menos” increase intra-household bargaining power?

1st UFPR Winter School on Economic Development

Aug 2023

Curitiba, Brazil

Can the school entrance age explain lower academic performance? Evidence from Peru.

LACEA Annual Meeting

Nov 2019

Puebla, Mexico

GRANTS, FELLOWSHIPS, HONORS AND AWARDS

Selected Participant, Price Theory Summer Camp, University of Chicago

Selective program on price theory, applying core economic tools to real-world problems

2026

Summer School Fellow, RIDGE

Selective summer program on frontier research in economics for outstanding graduate economists

2026

Annual Funding Competition - Global Affairs Canada/CIES

“Determinants of being a Peruvian criminal”. With Walter Noel, C\$5,000

2017

Think Tank Initiative - IDRC/GRADE

“Social interaction and crime: a regional crime analysis of Peru”. With Daniel Velasquez, US\$3,000

2016

SELECTED RESEARCH PROJECTS

Mobile Phone Expansion and Working Conditions in Peru: Difference-in-differences analysis exploiting telecom entry in 2006 to estimate impacts on employment quality and informality. (*Work in progress*)

Mismatch in the Egyptian labor market (with Caroline Krafft): Analyzes labor mismatch in Egypt, documenting structural and aspirational gaps, gender constraints, and pervasive over-education, with implications for job creation and education policy. (*Global Labor Organization Discussion Paper No. 1739*)

OTHER PUBLICATIONS

- * **Armas M., C.** and Escobal, J. (2018). Economic inclusion and territorial taxation: the case of high Andean tax exemptions.
- * **Armas M., C.** and Noel, W. (2017). Determinants of being a Peruvian criminal: social disorganization versus social learning.
- * **Armas M., C.** and Velazquez, D. (2017). Social interaction and crime: a regional crime analysis of Peru. 2017

TEACHING EXPERIENCE

Graduate Instructor, Mathematics Review Course
Department of Applied Economics, University of Minnesota

Summer 2024
Minneapolis, USA

Instructor (undergraduate level)
Pontifical Catholic University of Peru & San Martín de Porres University
Courses: Econometrics, Applied Public Economics, Stata, R, and Mathematics.

2013-2022
Lima, Peru

REFERENCES

- Caroline Krafft
- Christina Ewig
- Paul Glewwe
- Joe Ritter